

CMP-18 - Evidence Service

Admit and govern immutable Evidence Objects with provenance, custody, signatures, verification state, confidence, domain, and expiry.

Authority and state ownership

Book I: Decisions 24-29, 35, 40-43, 47-50, and 58. Book II: Chapters 18, 25. Authoritative state: Evidence Objects, custody chains, admission decisions, signature and verification state. Explicitly non-owned: Audit Records, telemetry, claims, Knowledge Graph, Trust Assessments.

Critical invariant

An admitted Evidence Object is immutable; correction occurs only through linked later objects.

Logical interfaces

CMP-18-IF-01 / Command / Submit Evidence Candidate

Command `Submit Evidence Candidate` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-18-IF-02 / Command / Admit Evidence Object

Command `Admit Evidence Object` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-18-IF-03 / Command / Challenge Evidence

Command `Challenge Evidence` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-18-IF-04 / Command / Record Supersession

Command `Record Supersession` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-18-IF-05 / Query / Get Evidence Object

Query `Get Evidence Object` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-18-IF-06 / Query / Trace Custody

Query `Trace Custody` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-18-IF-07 / Query / Verify Evidence Integrity

Query `Verify Evidence Integrity` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-18-IF-08 / Event / Evidence Admitted

Event `Evidence Admitted` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-18-IF-09 / Event / Evidence Challenged

Event `Evidence Challenged` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-18-IF-10 / Event / Evidence Superseded

Event `Evidence Superseded` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

Lifecycle, controls, and recovery

Lifecycle: Candidate > Admitted > Challenged > Superseded or Expired. All transitions require version, guard, identity, authority where applicable, idempotency, time context, and evidence. Contain: custody break, provenance forgery, mutable evidence, signature failure, unauthorized disclosure. Recovery proves identity, state, authority, provenance, dependencies, and invariants before mutation resumes.

Conformance tests

CMP-18-TST-001 - Sole-owner mutation. Attempt mutation through a consumer, replica, cache, operator path, and direct datastore route. Expected: Every unauthorized path is denied and evidenced.

CMP-18-TST-002 - Authority denial. Invoke every command with absent, expired, revoked, mismatched, and over-broad authority context. Expected: No state changes; denial reason and evidence are complete.

CMP-18-TST-003 - Valid lifecycle. Exercise every declared transition with valid guards and fresh versions. Expected: One transition Evidence Object is admitted per transition identifier; duplicate attempts produce no additional state mutation or admitted Evidence Object.

CMP-18-TST-004 - Invalid lifecycle. Attempt skipped, stale, replayed, and terminal-state transitions. Expected: Every invalid transition is rejected without partial mutation.

CMP-18-TST-005 - Critical invariant. An admitted Evidence Object is immutable; correction occurs only through linked later objects. Expected: Invariant holds under normal, concurrent, degraded, and recovery conditions.

CMP-18-TST-006 - Failure containment. Inject custody break, provenance forgery, mutable evidence, signature failure, unauthorized disclosure. Expected: Unsafe work stops, evidence is preserved, and declared containment state is observable.

CMP-18-TST-007 - Recovery. Restore from a verified checkpoint, reconcile dependencies, and resume after validation. Expected: Identity, state, authority, provenance, and evidence continuity are proven.

CMP-18-TST-008 - Contract compatibility. Run current and prior supported contract versions plus an unsupported version. Expected: Supported behavior remains compatible; unsupported input fails explicitly.

CMP-18-TST-009 - Privacy and security. Inject secrets, personal data, malicious input, confused-deputy requests, and log-exfiltration probes. Expected: Least privilege and data controls hold; prohibited data is absent from telemetry.

CMP-18-TST-010 - Topology independence. Repeat critical tests under permitted single-node, replicated, partitioned, and restored topologies. Expected: Topology does not multiply authority or mutation ownership.

Traceability and certification

Twelve requirements map to Books I-III and Book X. No open finding or Owner Review item. Final for Book IV v1.0.